

DermaForce

RAG + QLoRA for Hebrew Dermatology Board-Exam QA

Applied Language Models — Group Project B (solo)

The gap this project closes

A working RAG chatbot (Qdrant + OpenAI embeds + Claude) already existed as a product. Missing for the course:

- **Experiments & results** — no metrics, no ablations
- **Analysis** — no failure analysis, no hallucination study
- **QLoRA fine-tuning** — hard requirement, entirely absent
- Evaluating a model on **its own generated questions** (circular)

Fix: 949 real Israeli board-exam questions (2021–2026), each with an independent human-expert page/chapter citation into the exact corpus indexed.

System design

Hebrew MCQ → glyph fix → translate to EN → retrieve (EN corpus) → score 4 options
| |
└─ raw-Hebrew vs. translated query ─┘ (cross-lingual experiment)

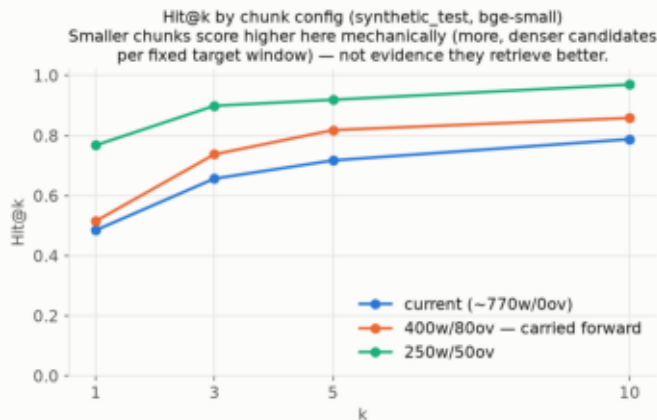
PRODUCT: Qdrant + OpenAI embeds → Claude → answer → verifier badge → Next.js UI
MEASUREMENT: chunks.jsonl → FAISS index variants → retrieval metrics
→ Claude-distilled QA → QLoRA Qwen2.5-3B → 4-arm eval

Two datasets

	Bologna + synthetic QA	Israeli board exams
Size	6,115 chunks → 1,219 QA items	949/950 MCQs, 7 sittings
Ground truth	Claude-generated (circular)	Human expert citation
Split	1,000/120/99, by chunk	Develop 2021-24 / hold out 2025+26
Role	Training data	The honest test set

Parsing required per-file format detection: 2 numbering conventions, 2 broken-glyph bugs, 3 reference formats, a 2023 cross-version alignment table, crop-box-intersected image detection.

Chunking fix



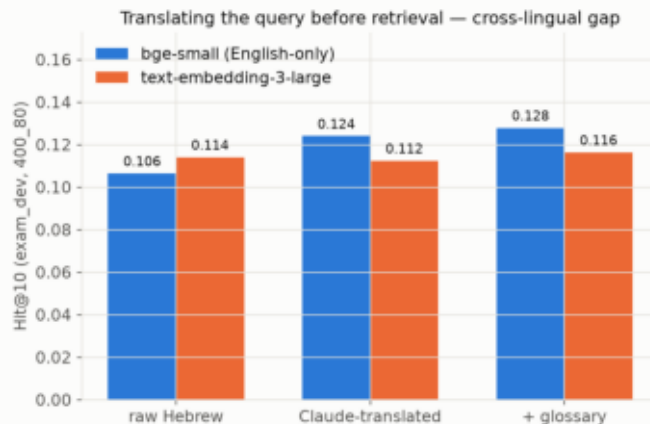
Old chunker: whole pages accumulated past a word target → 766w median, **zero** effective overlap. Fixed: exact word-stream window. 400w/80ov carried forward — not the top scorer, the honest one (see next slide).

Two eval sets pull chunk size in opposite directions

- **Synthetic gold** (page-overlap to one exact source chunk) → favors *more, smaller* chunks: denser candidates mechanically inflate Hit@k.
- **Exam gold** (± 2 -page tolerance around a candidate's own span) → favors *wider* chunks: more absolute page coverage per chunk, independent of content quality.

400_80 sits between both artifacts, never the worst on either — the config the chunking fix actually produces, not cherry-picked after the fact.

Cross-lingual retrieval — the headline result



Raw Hebrew vs. Claude-translated query, exam-dev, 400_80. bge-small (English-only) gains **+19% relative Hit@10** from translating first; text-embedding-3-large (real cross-lingual capability) stays flat.

Why translation helps — two worked examples

2024_09 Q57 (atopic dermatitis epidemiology):

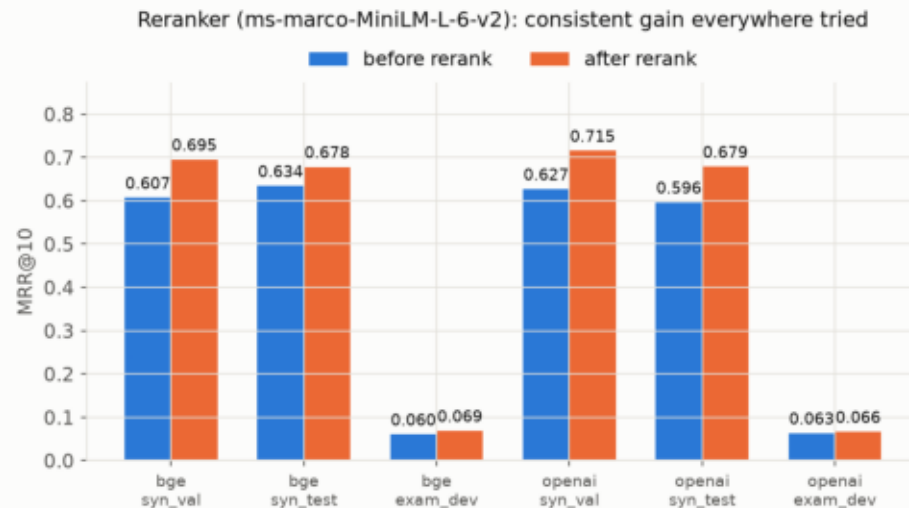
raw-Hebrew top-1 → *Fungal Diseases* (wrong) · translated top-1 → *Atopic Dermatitis* (right)

2024_09 Q79 (which nevus finding needs excision):

raw-Hebrew top-1 → *Erythema Multiforme/SJS/TEN* (wrong) · translated top-1 →
Benign Melanocytic Neoplasms (right)

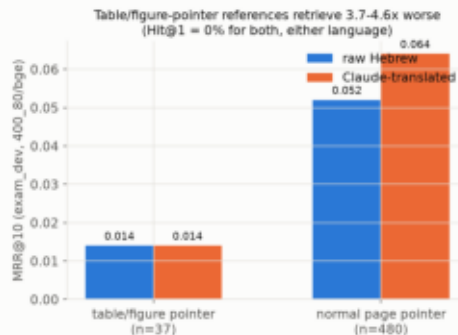
An English-only embedder has no lexical grip on untranslated Hebrew — translating restores literal term overlap it can use directly.

Reranker: a clean, low-risk win



ms-marco-MiniLM-L-6-v2 top-20 → top-10. Improves MRR@10 in **every one of 6** query-set × embedder combinations tested.

A distinct, worse retrieval-failure class



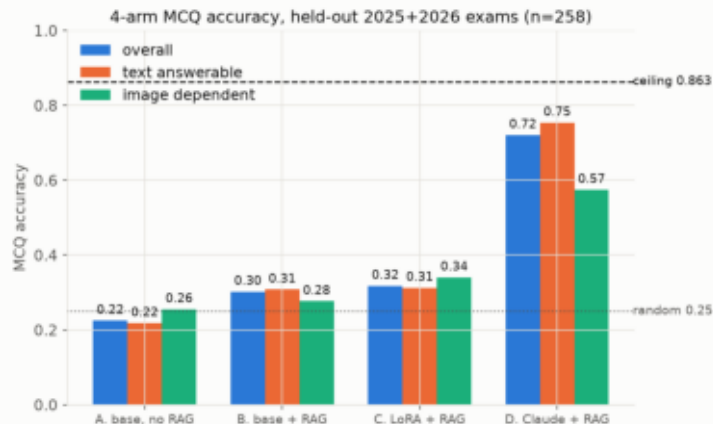
References pointing at a specific table/figure (37/517 dev questions) retrieve **3.7-4.6x worse**, Hit@1 = 0% in both languages — the direct cost of dropping figure captions at extraction time. A *retriever*-side ceiling, distinct from the *generator*-side image-dependent ceiling ahead.

QLoRA fine-tune

Qwen2.5-3B-Instruct, 4-bit NF4, LoRA rank $\in \{8, 16, 32\}$, trained on Colab.

- Target: citation-grounded answers, Claude-distilled, in the product's own [Chapter, p.N] format
- Training examples where the actual retriever misses the gold chunk $\rightarrow$ relabeled to explicit abstention (never taught to state facts absent from shown context)
- Val loss plateaus flat (~ 0.29 – 0.31) for all 3 ranks from step ~ 50 ; train loss drops fastest for $r=32$ — **overfitting scales with rank, measured not assumed**
- $r=16$ carried forward: least overfit rank with real capacity

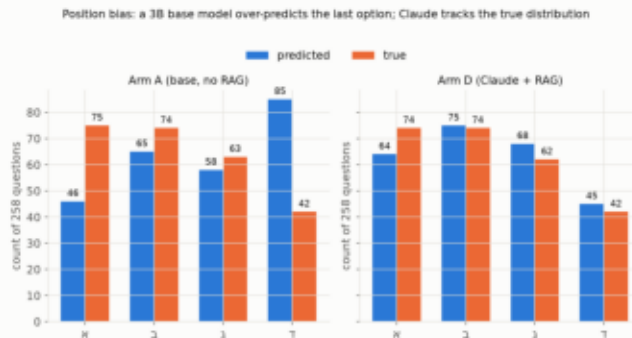
4-arm MCQ accuracy (n=258 held-out 2025+2026)



Ceiling = 86.3% (47/258 questions are image-dependent, unanswerable from text alone).

Claude+RAG captures **83.5% of everything a text-only system could get right** — not "72% accurate."

Position bias: model artifact vs. exam convention



Exam's own gold distribution is skewed (option A correct 75x, D only 42x — a real exam-writing convention). Arm A shows genuine *model* bias on top of that (over-predicts D 2x its true rate); Claude tracks the true distribution almost exactly.

Two artifacts caught before being reported as findings

- **Multi-hop references "score better"** (Hit@1 0.087 vs 0.028) — only because they resolve to a **1.7× larger gold-chunk set** on average. Not better multi-hop reasoning; an artifact.
- **97.3% of exam-dev questions overlap the synthetic set's sampled content** — stratified sampling already reaches everywhere. Rules out "coverage gap" as the explanation for the synthetic-vs-exam accuracy gap.

Discipline: every number here was checked against a second explanation before being written up as a finding.

Verifier agent — live demo moment

backend/verifier.py: extracts each claim + citation, checks it against the retrieved chunks, returns a groundedness score → UI badge (frontend/).

- Real, correctly-cited claim → groundedness_score = 1.0
- Fabricated citation ([Chapter: 12 Psoriasis, p.999]) attached to an unrelated claim → citation_valid = false, supported = false

Distinguishes real grounding from real hallucination — not just JSON shape.

Out-of-corpus abstention: 21/24 genuinely out-of-scope questions correctly declined (0.875 rate); the 3 that didn't abstain outright still scored groundedness 0.986 — nuanced partial answers, not confident hallucinations.

Limitations

- Translation is a demonstrated error source (though rarely fails outright)
- Page map is an interpolation (122/160 chapters anchored, rest inferred; ± 1 –2pp validated)
- Arm D (letter-generation) not directly comparable to A/B/C (log-likelihood)
- Two metric artifacts mean every Hit@k/MRR number here is "given this gold-set geometry," not absolute
- Both corpora copyrighted, not redistributed
- Image-dependent questions: a measured, hard ceiling for any text-only system

Conclusion & future work

A real product, measured honestly against **independent human ground truth**, not questions it generated itself — full retrieval ablation, 4-arm comparison, QLoRA fine-tune, and a verifier that demonstrably catches hallucination.

Next levers, by expected payoff:

1. Re-include figure captions as retrievable text (raises the ceiling, doesn't just measure it)
2. Deeper training-time retrieval budget for the LoRA arm (closer to product's top-8)
3. Larger multi-hop / table-figure eval slices, now cleanly isolated as failure classes

Thank you

Questions?